

PAVAN KUMAR ETTA

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PROFESSIONAL SUMMARY

M.S. graduate in Computer Information Science with **3.5+** years of Software Engineering experience building backend services, distributed systems, cloud infrastructure, and storage-backed applications using Python, Go, C++, AWS, Kubernetes, SQL, Redis, and object storage. Experienced in production deployments, data reliability, fault-tolerant system design, and performance optimization.

EDUCATION

University of Massachusetts, Dartmouth | *GPA: 4.0*

North Dartmouth, USA

M.S. in Computer Information Science — Data Science

Aug 2024 - May 2026

Indian Institute of Information Technology | *GPA: 3.2*

SriCity, INDIA

B.Tech in Electronics and Communication Engineering

Aug 2017 - June 2021

WORK EXPERIENCE

Graduate Research Assistant | **University of Massachusetts, Dartmouth** | USA

Mar 2025 – Dec 2025

- Architected a **Next.js-based** web application with a **Flask** backend and **MongoDB**, enabling users to upload financial documents and view detailed reports generated through an **NLP** pipeline.
- Integrated **LangChain** for AI workflow orchestration and optimized ChatGPT-based text extraction, improving extraction accuracy by **30%**; implemented BERT-based classification with **93%** accuracy.

Software Engineer II | **Rhenix Lifesciences LLP** | India

Aug 2021– Aug 2024

- Built **iOS**, **Android**, and web features for the CureOrg platform, enabling patients to link **5+** wearables and view daily heart rate, sleep, steps, calories, activity, distance, and elevation metrics through Fitbit, Garmin, Withings, and Apple Health integrations.
- Automated daily health-data retrieval, reducing collection effort by **~90%**, by building Python ingestion jobs and Django commands on **EC2** to manage authentication, refresh tokens, parse JSON, retry failed requests, and store validated records.
- Improved research-data reliability by converting **5-second heart-rate** signals into **1-minute windows** using SQL, validating API data, handling missing records, and storing metrics in **MySQL** on **AWS EC2**.
- Reduced sync debugging time by **~40%** by building internal monitoring views with validation logs and **AWS CloudWatch** alerts to identify failed API calls, delayed jobs, and missing records faster.
- Improved health-metrics dashboard performance by up to **70%** by optimizing **Django REST API** responses, MySQL query patterns, indexing, pagination, and **Redis caching** for frequently accessed user metrics.
- Accelerated backend and dashboard deployments by **80%** by containerizing services with Docker and automating production deployments on **AWS EKS** using GitLab CI/CD pipelines.

Software Engineer Intern | **Cure Science Institute** | USA

Jun 2021 – Aug 2021

- Developed a **BLE-based** asset tracking system using an Android app, BLE beacons, gateways, and REST APIs, enabling real-time location data uploads to **AWS S3** with persistent background services for **24/7** tracking and improved battery efficiency.
- Built an admin analytics dashboard using React, **Google Maps**, AWS API Gateway, **DynamoDB & RDS**, CloudWatch, and scheduled jobs to monitor device health, visualize coordinates, detect offline gateways, and analyze asset movement patterns.

SKILLS

Programming: Python, JavaScript, TypeScript, C/C++, Java, Go

Backend & APIs: Django, Flask, FastAPI, REST APIs, gRPC, GraphQL, Event-Driven Architecture, System Design, Postman

Web Tools: jQuery, Angular, Node.js, Vue.js, Next.js, Flutter, Nginx, Figma, HTTPS

Databases & Caching: PostgreSQL, MySQL, DynamoDB, MongoDB, Redis, Elasticsearch

Cloud & Infra: AWS (Lambda, EC2, S3, RDS, MSK, EKS), GCP, Azure, Terraform, Serverless

Data Engineering: Apache Spark, PySpark, Airflow, Snowflake, Hadoop, dbt

DevOps & Observability: Docker, Kubernetes, CI/CD, GitHub Actions, GitLab CI, Jenkins, Prometheus, Grafana

PROJECTS

Trip Planner [\[GITHUB\]](#)

(React.js, TypeScript, Go, DynamoDB, React Native, AWS)

- Built and deployed a full-stack trip-planning platform using **React**, TypeScript, Go, DynamoDB, **AWS Amplify**, and **Google Places API** to generate personalized itineraries and sync them with a React Native mobile app for on-the-go access.

Cloud-Native API Gateway & Rate Limiter [\[GITHUB\]](#)

(Python, Go, AWS, Redis, Docker, Kubernetes)

- Designed a **cloud-native API** gateway with authentication, request routing, rate limiting, retries, circuit breakers, and service-level observability, improving backend reliability and protecting microservices from traffic spikes.

Fault-Tolerant Sharded Key-Value Store [\[GITHUB\]](#)

(C++, Raft, SQLite)

- Engineered a fault-tolerant sharded key-value store in **C++** using **Raft** consensus and SQLite, implementing replicated state machines across multiple servers to maintain consistency, support failure recovery, and achieve **94.99%** availability.

Wandermate [\[GITHUB\]](#)

(React, TypeScript, Node.js, MongoDB, Socket.IO)

- Developed a full-stack travel platform using React, TypeScript, Node.js, and MongoDB, integrating **Socket.IO** for real-time messaging, **PropelAuth** for secure authentication, **Mapbox** for interactive maps, and reusable APIs for third-party integrations.